To B, Still in A

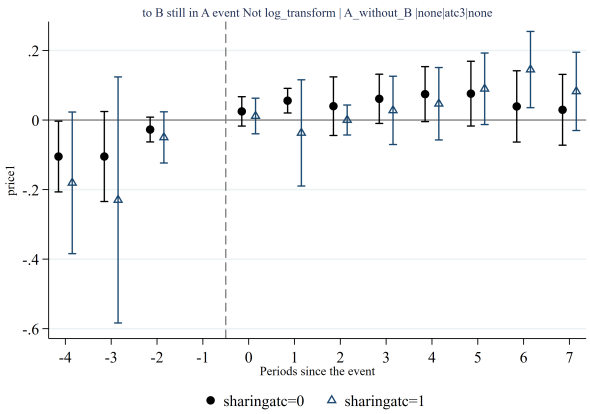
Comparison of `csdid` and `did_imputation`

BoardPharma

November 11, 2026

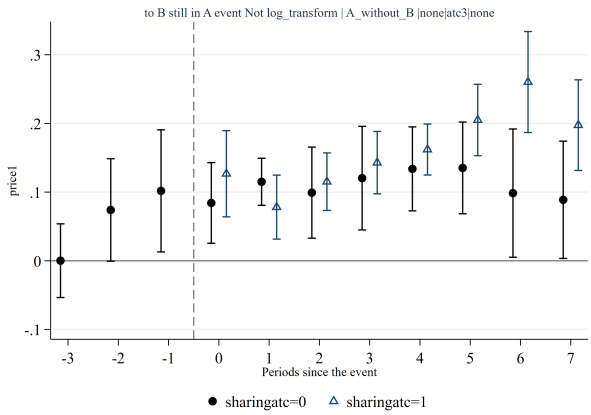
1 Event-study figures

A side, `csdid`



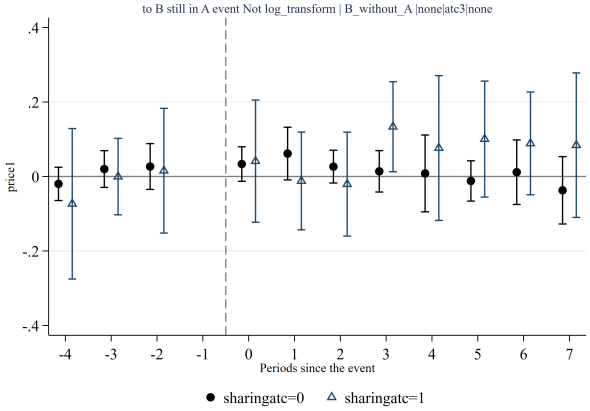
Estimation observations: 108,156 (share 0), 28,428 (share 1)

A side, `did_imputation`



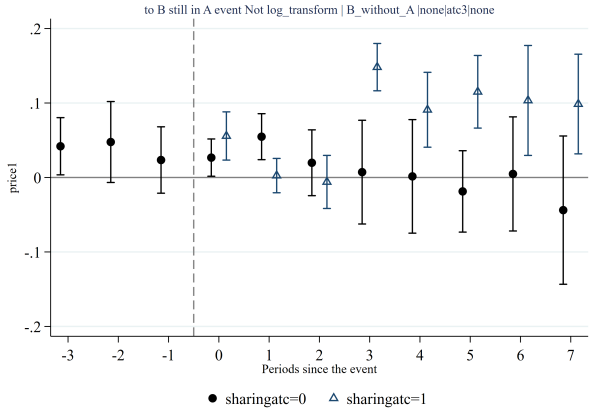
Estimation observations: 108,396

B side, `csdid`



Estimation observations: 126,504 (share 0), 21,696 (share 1)

B side, `did_imputation`



Estimation observations: 130,236

Notes. The four figures are the original Stata exports for `to_B_still_in_A`, with ruleout off and ATC3 sharing fixed at the event quarter (`dynamic_share=0`). Black circles and hollow blue triangles identify share 0 and share 1. Error bars are 95% confidence intervals. The figures retain their original vertical scales. The outcome is log `price1` after upper-p95 winsorization. Both methods use req2, the Not control group, and company clustering. `csdid` fits each sharing group separately within the cohort stacks and uses quarter −1 as the reference. `did_imputation` uses company–product–cohort and cohort–calendar-quarter fixed effects and reports sharing-specific post-event estimates from its `hetby` run. The `csdid` counts are per fit and can include the same control observations in both counts.

2 Quarterly coefficients

A side

Pre-event coefficients

	csdid		did_imputation	Difference vs. pooled lead	
Quarter	Share 0	Share 1	Pooled	Share 0	Share 1
−4	−0.104973	−0.180621	—	—	—
−3	−0.104902	−0.229829	0.000117	−0.105019	−0.229946
−2	−0.027290	−0.049903	0.073995	−0.101285	−0.123898
−1	0 (ref.)	0 (ref.)	0.101788	—	—

Event and post-event coefficients

	Share 0			Share 1		
Quarter	csdid	did_imputation	Difference	csdid	did_imputation	Difference
0	0.024882	0.084173	−0.059291	0.011651	0.126739	−0.115088
1	0.055719	0.115011	−0.059291	−0.037006	0.078082	−0.115088
2	0.039874	0.099165	−0.059291	0.000094	0.115182	−0.115088
3	0.061037	0.120328	−0.059291	0.027790	0.142879	−0.115088
4	0.074534	0.133826	−0.059291	0.046907	0.161995	−0.115088
5	0.075913	0.135205	−0.059291	0.089897	0.204985	−0.115088
6	0.039213	0.098504	−0.059291	0.145206	0.260294	−0.115088
7	0.029485	0.088776	−0.059291	0.082432	0.197520	−0.115088

B side

Pre-event coefficients

	csdid		did_imputation	Difference vs. pooled lead	
Quarter	Share 0	Share 1	Pooled	Share 0	Share 1
−4	−0.019846	−0.073401	—	—	—
−3	0.019990	−0.000284	0.041940	−0.021949	−0.042224
−2	0.026729	0.015538	0.047626	−0.020897	−0.032088
−1	0 (ref.)	0 (ref.)	0.023465	—	—

Event and post-event coefficients

	Share 0			Share 1		
Quarter	csdid	did_imputation	Difference	csdid	did_imputation	Difference
0	0.033398	0.026680	0.006718	0.041192	0.055730	−0.014537
1	0.061490	0.054772	0.006718	−0.012008	0.002529	−0.014537
2	0.026479	0.019761	0.006718	−0.020521	−0.005983	−0.014537
3	0.013886	0.007168	0.006718	0.133631	0.148169	−0.014537
4	0.008187	0.001469	0.006718	0.076379	0.090916	−0.014537
5	−0.011984	−0.018702	0.006718	0.100522	0.115060	−0.014537
6	0.011482	0.004764	0.006718	0.088885	0.103422	−0.014537
7	−0.037119	−0.043837	0.006718	0.084085	0.098623	−0.014537

Notes. Differences are `csdid` minus `did_imputation`, in log points. The pre-event subtractions compare estimates with different reference periods and pooling rules. The `did_imputation` leads are pooled across both sharing groups and estimated separately from the post-event effects. The `csdid` zero at −1 is the reference. A dash denotes an unavailable estimate or comparison.

3 The post-event shift and the pre-trend

Within each side and sharing group, the post-event gap is constant across quarters 0 through 7 to the precision of the saved estimates. For share 1, the `csdid` curve is lower by 0.115088 log points on side A and by 0.014537 on side B. Its quarter-to-quarter changes are the same as those of `did_imputation`.

Table 1: Fixed post-event differences and `csdid` pre-event tests

Side	Sharing group	Post-event difference	Joint lead test p-value
A	0	−0.059291	0.203
A	1	−0.115088	0.353
B	0	0.006718	0.337
B	1	−0.014537	0.090

Notes. The joint Wald test uses the three `csdid` leads (−4, −3, and −2), relative to quarter −1. The statistic is $\hat{b}'\hat{V}^{-1}\hat{b}$, using the company-clustered covariance matrix in the CSV. The reported p-values use an asymptotic $\chi^2(3)$ reference distribution.

In the simple case where both methods use the same units and weights and differ only in their reference, `csdid` subtracts the treated-minus-control log price gap at quarter −1. The imputation estimate subtracts the average pre-event gap. The difference between their post-event coefficients is therefore

$$\text{average pre-event gap} - \text{quarter } -1 \text{ gap}.$$

Under this interpretation, the downward shifts for share 1 indicate that the quarter −1 price gap was larger than its pre-event average on both sides. The saved runs use staggered treatment timing and different estimation setups. Attributing the full shift to the reference requires a comparison with the same units and weights.

On side A, all three share 1 `csdid` leads are negative relative to −1. On side B, the earlier leads are mixed. The joint tests for share 1 give $p = 0.353$ on A and $p = 0.090$ on B. The tests do not reject at the 5% level on either side. The estimates leave room for a pre-existing change in the price gap, or for sampling variation in the last pre-event quarter. A larger gap at −1 alone cannot establish a steady trend or the direction of either group's own prices.

Sources. The four original Stata PNGs and their coefficient CSVs are from the A- and B-side `to_B_still_in_A` folders (`req2`, `price1`) for the ruleout-off specification. The benchmark comparison of reference periods follows Roth, Sections 3.2 and 3.3.